# Lab 1: Requirements Engineering & UML Use-Case Modelling

*Problem Statement #55 — Book Club Reading Challenge & Discussion Portal*

## 1. Problem Context

A community reading portal that tracks personal reading page goals, facilitates spoiler-tagged chapter discussion threads, and runs monthly voting polls for next book selection. Stakeholders: Club Member, Discussion Lead.

## 1. Requirements Table

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall hide spoiler-tagged discussion content until the club member explicitly reveals it or marks the corresponding chapter as read. | High | Pass: Spoiler content stays hidden until the member reveals it or marks the chapter read. Fail: Spoiler content is visible before either action. | Prevents members from accidentally viewing content from unread chapters. |
| FR-002 | Functional | The system shall allow a club member to create and participate in chapter-specific discussion threads. | High | Pass: A member can create a discussion tied to a selected chapter and submit a reply. Fail: The discussion cannot be tied to a chapter, or the reply cannot be submitted. | Enables organized discussion of individual book chapters. |
| FR-003 | Functional | The system shall allow a club member to set and track a personal reading page goal for a selected book. | High | Pass: The member can enter a page goal and the system displays current progress toward it. Fail: The goal cannot be saved or progress cannot be displayed. | Helps members monitor their individual reading progress. |
| FR-004 | Functional | The system shall allow a club member to cast one vote in an active monthly poll for selecting the next book. | High | Pass: An eligible member can select one book and submit one vote during an active poll. Fail: The system accepts more than one vote from the same member. | Supports fair monthly selection of the club's next book. |
| FR-005 | Functional | The system shall allow the Discussion Lead to create, manage, and close a monthly book-selection poll. | Medium | Pass: The Discussion Lead can create a poll, add book choices, and close it. Fail: Any required poll-management action cannot be completed. | Allows the Discussion Lead to manage the monthly book-selection process. |
| NFR-001 | Nonfunctional | The system shall prevent duplicate votes using user session verification and update monthly poll tallies in real time. | High | Pass: Testing confirms duplicate votes are rejected and valid vote tallies update under simulated peak load. Fail: A duplicate vote is accepted, or tally updates fail. | Protects voting integrity and provides timely poll results. |
| NFR-002 | Nonfunctional | The system shall restrict discussion content and member information to authorized, authenticated users only. | High | Pass: An unauthorized user attempting to access protected discussion or member information is denied access. Fail: Protected information is shown to an unauthorized user. | Protects member information and discussion content from unauthorized access. |

## 2. Actors and Use Cases

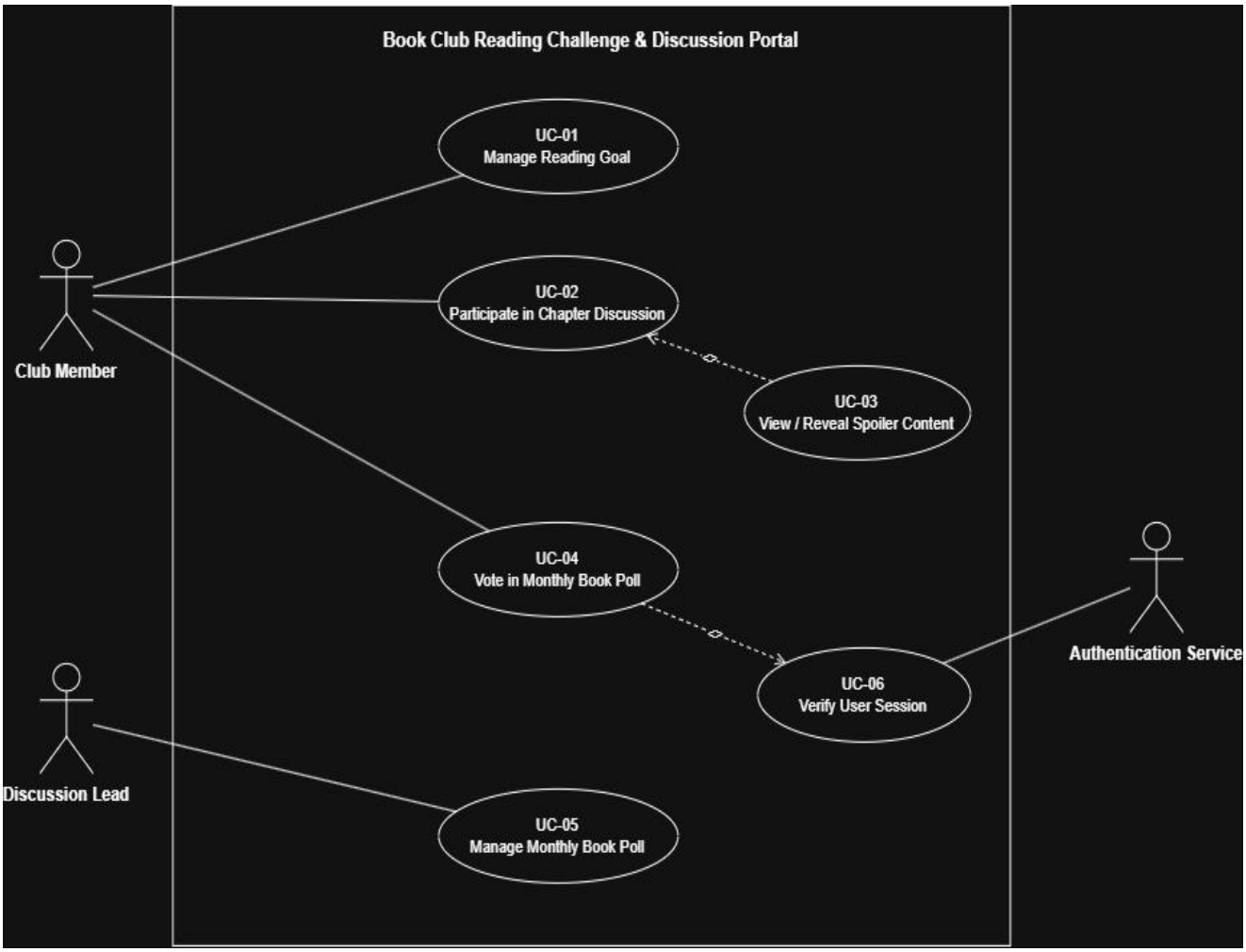
**Actors:**

- Club Member — sets reading goals, participates in chapter discussions, reveals spoiler content, and votes in the monthly poll.
- Discussion Lead — creates, manages, and closes the monthly book-selection poll.
- Authentication Service — verifies the member's session before protected actions such as voting.

**Use Cases:**

- UC-01 Manage Reading Goal
- UC-02 Participate in Chapter Discussion
- UC-03 View / Reveal Spoiler Content — «extend» of UC-02
- UC-04 Vote in Monthly Book Poll — «include» Verify User Session
- UC-05 Manage Monthly Book Poll
- UC-06 Verify User Session

## 3. UML Use-Case Diagram



## 4. Use-Case Flow Specification

**Use Case: UC-04 — Vote in Monthly Book Poll**

Primary Actor: Club Member

**Preconditions:**

1. The club member has an active, verified user session.
2. A monthly book-selection poll is currently open.
3. The club member has not already voted in the current poll.

**Postconditions:**

4. The member's vote is recorded and cannot be submitted again for the same poll.
5. The poll tally is updated and a confirmation is displayed.

**Main Success Scenario:**

6. Club Member opens the monthly book-selection poll.
7. System verifies the member's session (includes UC-06 Verify User Session).
8. System displays the available book choices.
9. Club Member selects one book and submits the vote.
10. System checks that the member has not already voted in the current poll.
11. System records the vote and updates the poll tally.
12. System displays a confirmation message. Use case ends successfully.

**Alternate Flow — 5a. Duplicate Vote Attempt:**

13. System detects that the member has already voted in the current poll.
14. System rejects the new vote and displays a message that a vote was already cast.
15. The existing vote remains unchanged. Use case ends without recording another vote.